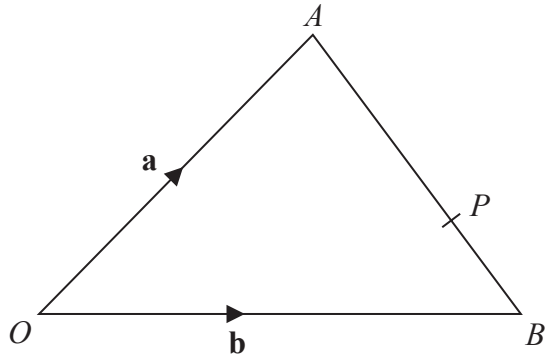
19 OAB is a triangle.



$\overrightarrow{OA} = \mathbf{a}$
$\overrightarrow{OB} = \mathbf{b}$
P is the point on AB such that $AP : PB = 3 : 2$

Find $\overrightarrow{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$AB = b - a$

1 mark
AB = b − a.

$AP = 3(b - a) \div 5$

$OP = OA + AP$

$= a + 3(b - a) \div 5$

1 mark
Use 3/5 of AB and add OA.

$= (5a + 3b - 3a) \div 5$

$OP = \dfrac{2a + 3b}{5}$

1 mark
Collect vector coefficients.

(Total for Question 19 is 3 marks)